

# **Cognitive Psychology and Neuropsychology: Study & Revision Notes**

# 1. Foundations, History, and the Nature vs. Nurture Interplay

Psychology is the scientific study of the mind, the brain, and how they drive our behavior. For a long time, people argued about "nature versus nurture"—asking whether we are purely shaped by our biology or born as blank slates shaped by our surroundings. Today, cognitive psychology looks at this not as a battle, but as a continuous interaction. Your genes build possibilities, but your environment shapes how those genes show up in real life.

## Key Points to Know

- **Genotype vs. Phenotype:** Your *genotype* is the exact DNA code you inherit from your parents. Your *phenotype* is what actually gets expressed—your physical traits, your personality quirks, and how you behave.
- **Heritability:** This is a statistical estimate showing how much the differences between people in a specific population are driven by genetic differences rather than lifestyle or surroundings.
- **Twin and Adoption Studies:** Identical (monozygotic) twins share 100% of their genes, while fraternal (dizygotic) twins share about 50%. If identical twins raised apart show the same traits, that points to genetics; if adopted kids resemble their adoptive parents more than biological ones, that highlights the environment.
- **Shared vs. Non-Shared Environments:** Understanding how common household factors compare to unique personal experiences in shaping individual traits.
  - *Shared environment:* Things siblings experience together, like household income, parenting styles, family diet, and the neighborhood.
  - *Non-shared environment:* Unique individual experiences, like different friend groups, teachers, or personal life events. Research shows non-shared factors often have a stronger impact on making siblings different from each other.
- **Four Gene-Environment Interactions:** Key ways genetic tendencies correlate with environmental influences.
  - *Assortative Mating:* People tend to choose partners who have similar physical traits or personalities rather than mating totally at random.
  - *Passive Correlation:* Parents pass on their genes and also create the home environment (for instance, musical parents passing on musical genes while filling the house with instruments).
  - *Evocative (Reactive) Correlation:* A person's inherited traits trigger specific reactions from the people around them (a child who hums all the time gets noticed and sent to music classes).
  - *Active Correlation:* People actively choose environments that match their genetic tendencies (a teenager with musical talent seeks out band practice on their own).
- **Early Roots and Main Schools:**
  - *Ancient Roots:* Aristotle thought the mind was centered in the heart, while Plato linked it to the brain. Later, in 900 CE, the physician Avicenna pioneered early work on memory loss and psychiatric symptoms.

- *Structuralism (Wilhelm Wundt)*: Treated psychology as an experimental science in a lab, trying to break down conscious thought into tiny parts using trained introspection.
- *Functionalism (William James)*: Argued that the mind is too dynamic to dissect into tiny pieces; instead, we should study *why* and *how* the mind helps us adapt and function.
- *Psychoanalysis (Sigmund Freud)*: Suggested our conscious mind is only the surface, and that behavior is largely driven by unconscious conflicts and early childhood events.
- **Mind-Body Debates**: Core philosophical and scientific perspectives on how the mind relates to the physical brain.
  - *Dualism*: The idea that the physical body and the non-physical mind are two separate entities.
  - *Mind-Brain Reductionism*: The view that mental experiences can be explained by physical brain activity. The challenge here is explaining how raw electrical signals turn into rich, subjective thoughts and feelings.

## Everyday Examples

- **Gene-Environment Loop**: Think of a tall kid with great coordination. Because of those genes, coaches encourage them to play basketball (evocative correlation), and the kid willingly spends weekends on the court (active correlation). By adulthood, their athletic skill is a blend of biology and practice.
- **The Computer Metaphor for the Mind**: Think of the brain as the hardware (circuits and chips) and the mind as the software (operating system and apps). You cannot understand how an app functions just by staring at the silicon chip.

## Key Takeaways

- Nature and nurture work as an interconnected system, not two opposing forces.
- Twin and adoption studies help separate genetic influences from environmental effects.
- Your genes influence the environments you end up experiencing.
- Structuralism tried to break the mind into small components, while Functionalism asked what purpose mental processes serve.

# 2. How We Study the Brain: Methods and Neuroimaging

Because we cannot crack open a healthy person's head just to see what they are thinking, cognitive psychologists use clever indirect tools. We look at what happens when parts break down (lesion studies), we record brain signals from the outside, and we scan the brain while people perform specific tasks.

## Key Points to Know

- **Four Core Ways to Study the Nervous System**: Primary experimental approaches used

in modern neuroscience.

1. *Examine brain damage (lesion studies)*: See what ability is lost when an area is injured.
  2. *Record brain activity*: Measure electrical, magnetic, or metabolic activity during tasks.
  3. *Correlate brain anatomy with behavior*: Compare brain structure sizes to cognitive skills.
  4. *Stimulate brain tissue*: Artificially activate or temporarily pause a brain area.
- **Neuropsychology and Lesions**: Insights and limitations gained from observing brain damage.
    - Helps identify what brain regions do by linking damaged areas to missing skills (like speech loss after Broca's area damage).
    - *Major limitation*: Brain damage is rarely neat or limited to one spot. We also rarely have a "baseline" measure of how the person performed before the injury.
    - *Brain Plasticity*: The brain reorganizes itself after injury, meaning healthy areas can take over lost jobs and make tests tricky to interpret.
  - **Functional Neuroimaging (Measuring Brain Activity)**: Techniques for tracking real-time neural function and metabolic processes.
    - **fMRI (Functional Magnetic Resonance Imaging)**: Tracks changes in blood flow and oxygen levels. When neurons work, they consume oxygen, triggering an influx of fresh oxygenated blood—known as the **BOLD effect** (Blood Oxygenation Level-Dependent). It provides high spatial resolution (sharp 3D pictures made of "voxels") but low temporal resolution because blood moves much slower than electrical thoughts.
    - **PET (Positron Emission Tomography)**: Injects a tiny, harmless radioactive tracer to track glucose metabolism and blood flow. Good at showing activity levels, but expensive, slow, and involves radiation.
    - **EEG (Electroencephalography)**: Places electrodes on the scalp to pick up the electrical firing of large groups of neurons. It has temporal resolution (measures things down to the millisecond) but poor spatial resolution because the signal blurs through the skull.
    - **MEG (Magnetoencephalography)**: Measures the magnetic fields created by neural electricity. Fast temporal tracking like EEG, with better tracking of where signals travel, but very costly.
  - **Structural Imaging (Looking at Brain Shape)**: Tools for examining anatomical structures and tissue density.
    - **CT Scan**: Uses multiple X-ray angles to show structural density (bone, tissue, tumors), but exposes the patient to radiation and does not show active function.
    - **MRI**: Uses strong magnets and radio waves to wiggle hydrogen atoms (protons) in our body's water. Produces sharp anatomical images without radiation.
  - **Brain Stimulation**: Techniques for non-invasively probing localized brain function.
    - **TMS (Transcranial Magnetic Stimulation)**: Holds a magnetic wand against the skull to temporarily stimulate or switch off a tiny patch of neurons. It lets researchers run "cause-and-effect" experiments safely without causing permanent damage.

## Everyday Examples

- **Spatial vs. Temporal Resolution**: Imagine tracking a car race. An fMRI is like a high-resolution camera with a slow shutter speed—it gives a crisp, detailed photo, but

you cannot tell exactly which millisecond the car crossed the line. An EEG is like a stopwatch—it tells you the exact millisecond the car passed, but the photo is too blurry to see the driver's face.

- **TMS as a Virtual Switch:** If you place a TMS wand over the motor area controlling the hand, a quick magnetic pulse can cause a participant's finger to twitch involuntarily, proving that specific patch controls that physical movement.

## Key Takeaways

- No single neuroimaging tool does everything; researchers pick based on whether they need to know *where* (spatial resolution) or *when* (temporal resolution).
- fMRI measures blood oxygen changes (the BOLD response), not direct electrical firing.
- EEG and MEG track electrical and magnetic signals in real time.
- Brain plasticity means healthy brain areas can rewire to take over lost functions after an injury.

### 3. Human Language: How We Learn to Talk and Understand

Babies learn to talk long before they learn formal rules. They go from hearing a continuous, blurry stream of sound to breaking it into words, building sentences, and understanding complex social meaning.

#### Key Points to Know

- **Speech Perception in Babies:** How infant auditory systems process phonetic boundaries over time.
  - Young infants (under 6–12 months) can distinguish almost any phoneme (sound unit) across world languages.
  - After about 12 months, the brain tunes out sounds not used in their native environment, focusing entirely on their home language.
  - *Phonotactic Rules:* Rules about which sounds can go together in a specific language (for instance, English words never start with "ng").
- **Statistical Learning vs. Universal Grammar:** The debate between empirical pattern detection and innate language acquisition structures.
  - *Statistical Learning Hypothesis:* Babies act like little data analysts. They calculate how often syllables appear side-by-side to guess where one word ends and the next begins.
  - *Universal Grammar (Noam Chomsky):* Argues that the speech children hear is too messy, limited, and full of baby talk (the "poverty of the stimulus") for them to learn complex grammar purely by statistics. Therefore, kids must have an innate, biological blueprint for grammar.
- **Stages of Speech and Syntax Development:** The chronological progression of early expressive language capacity.
  1. *Babbling:* Starts with vowels and moves to simple consonants made with easy jaw movements (like "ba-ba", "ma-ma").
  2. *Telegraphic Stage (12–26 months):* Two-word sentences missing filler words (e.g., "doggy go").
  3. *Grammatical Morphemes (27–30 months):* Kids add plurals, prepositions, questions, and negations (e.g., changing "no want milk" to "don't want milk").
  4. *Complex Sentences (35–46 months):* Children start embedding clauses and combining multiple ideas (e.g., "the boy who hit me is coming").
- **Morphology and Overregularization:** How children extract and apply linguistic rules across vocabulary acquisition.
  - We don't memorize every single word; we memorize word roots and apply mental rules (like adding "-ed" for the past tense).
  - When kids learn a rule, they often make **overgeneralization errors** (saying "goed" instead of "went", or "swimmed" instead of "swam"). This proves children are creatively applying internal rules, not just mimicking adults.

- **Semantics and the Extension Problem:** Mechanisms involved in mapping word labels to real-world objects and concepts.
  - How does a toddler know what a word refers to?
  - *Overextension:* Using one word too broadly (calling all furry animals "doggy" or all cats "Garfield").
  - *Underextension:* Using a word too narrowly (thinking "cat" only refers to the specific family pet at home).
  - *Social Pragmatic Account:* Kids solve this through **joint attention**—looking at what an adult is pointing at and understanding the adult's intention to name that object.

## Everyday Examples

- **Statistical Learning in Action:** If a baby repeatedly hears the phrase "pretty baby", the syllables *pre-tty* and *ba-by* always appear together, but *tty-ba* only occurs sometimes. The baby's brain notes this statistical pattern and deduces that "pretty" and "baby" are two separate words.
- **The "Goed" Mistake:** When a 3-year-old says "I goed outside", parents rarely correct them with a grammar lecture. The child still figures out the irregular exception ("went") over time through continued exposure.

## Key Takeaways

- Babies start as universal listeners and narrow their focus to native language sounds by age one.
- Overgeneralization errors ("swimmed") prove that children build internal rule systems rather than just copying what they hear.
- Language acquisition relies on both statistical pattern tracking and social cues like joint attention.
- Sentences get their meaning from syntax (word order), not just the vocabulary words themselves.

# 4. Working Memory and Sensory Perception

Memory is not a single storage box, and perception is not just taking photos of the outside world. Our brain actively manipulates information in real time while translating raw sensory signals (light and air pressure) into meaningful objects and sounds.

## Key Points to Know

- **Working Memory vs. Short-Term Memory:** Key functional differences between passive retention and active processing.
  - *Short-term memory (STM):* Simple passive holding of information for a few seconds.
  - *Working memory (WM):* Actively holding, updating, and manipulating information to solve problems, calculate, or make sense of a sentence.
- **The Fall of the "Bottleneck" Model (Patient KF):** Clinical evidence disproving single-channel sequential memory processing.
  - Old models claimed all long-term memories had to pass through a single short-term

bottleneck.

- *Patient KF* suffered brain damage and could only hold 1 or 2 spoken numbers in short-term memory (normal capacity is about 7), yet his long-term memory was completely intact. This proved short-term and long-term memory use different neural systems, and verbal information is processed separately from visual information.
- **Baddeley & Hitch Working Memory Model:** The multi-component model explaining parallel domain-specific processing subsystems.
  - **Central Executive:** The supervisor that directs attention, coordinates tasks, and handles problem-solving.
  - **Phonological Loop:** Deals with spoken and written words. It has a short-term store and an articulatory rehearsal process (your "inner voice" repeating words so you do not forget them). It has a limited capacity, which is why longer multi-syllable words are harder to remember (the *word-length effect*).
  - **Visuo-Spatial Sketchpad:** Your "inner eye" that manipulates mental pictures, plans spatial routes, and keeps you from bumping into furniture.
- **Auditory Perception (Hearing):** Mechanisms for converting acoustic waves into organized neural signals.
  - Sound is created by air pressure waves vibrating the eardrum and the tiny bones of the middle ear (the ossicular chain).
  - Vibrations enter the fluid-filled **cochlea**, bending tiny hair cells on the basilar membrane to trigger electrical nerve impulses.
  - **Tonotopic Map:** Both the cochlea and the primary auditory cortex are laid out like a piano keyboard—neurons at one end respond best to low pitches, and the other end responds to high pitches.
  - *Music vs. Language:* They have a double dissociation. Brain damage can cause **amusia** (loss of music perception) while speech remains fine, or **aphasia** (loss of language) while music skills stay intact.
  - *Auditory Agnosia:* Hearing is fine, but the brain cannot interpret what the sound means (verbal auditory agnosia means spoken words sound like a continuous buzz).
- **Visual Perception (Seeing):** Pathways for converting light energy into spatial and object representations.
  - Light hits the **retina** at the back of the eye, where photoreceptor cells turn it into electrical signals.
    - **Rods:** Highly sensitive to dim light; handle black-and-white night vision.
    - **Cones:** Need bright daylight; detect color (red, green, blue) and fine detail.
  - Visual signals cross at the optic chiasm and travel to the **primary visual cortex** in the occipital lobe, which organizes space using a **retinotopic map**.
  - **Two Visual Streams:** Anatomically and functionally distinct visual processing pathways.
    - **Ventral Stream ("What" pathway):** Travels downward into the temporal lobe to identify objects, colors, and faces. Damage here causes **visual agnosia** (inability to identify seen objects) or **prosopagnosia** (face blindness).
    - **Dorsal Stream ("Where / How" pathway):** Travels upward into the parietal lobe to track location, motion, and guide motor actions (like reaching out to grab a glass).

- **Blindsight:** Damage to the primary visual cortex can leave a person consciously blind, yet they can still guess the location of moving objects because subcortical visual pathways remain active.

## Everyday Examples

- **Dual-Task Processing:** You can easily walk (spatial task) while listening to a podcast (verbal task) because they use separate working memory subsystems. But try reading a complex email while someone reads out a list of phone numbers—both fight for the phonological loop, and performance crashes.
- **The Ventral Stream Failure:** In *The Man Who Mistook His Wife for a Hat*, a patient looked at his wife and grabbed her head trying to put it on. His eyes could see shapes, but his ventral stream could not match the visual shape to the concept of a human face.

## Key Takeaways

- Working memory handles active manipulation of information, not just passive storage.
- Visual and verbal working memory run on separate tracks and can operate in parallel.
- The auditory and visual systems preserve physical maps of the world (tonotopic for sound frequency, retinotopic for visual space).
- Vision splits into two core highways: the ventral "what" stream (identifying objects) and the dorsal "where/how" stream (guiding action).

## 5. Attention and Consciousness: How We Experience the World

Your senses take in massive amounts of data every second, but your conscious brain can only process a tiny fraction of it. Attention is the gateway that selects what gets processed in high detail and what gets ignored.

### Key Points to Know

- **Types of Attention:** The distinct modes through which processing capacity is allocated.
  - *Voluntary (Top-Down):* Goal-driven, conscious focus (like searching a crowded diagram for a specific icon).
  - *Involuntary (Bottom-Up):* Stimulus-driven focus (a loud bang or a flashing red light grabs your attention instantly).
  - *Selective Attention:* Focusing on one conversation while filtering out background noise.
  - *Sustained Attention:* Keeping your focus locked on a single task over time.
  - *Divided Attention (Multitasking):* Splitting resources across tasks, which typically degrades performance on both.
  - *Effortless Attention (Flow):* High focus without feeling mentally drained, common in skilled musicians or athletes.
- **The Spotlight and Feature Integration:** Theoretical models explaining spatial orientation and feature binding.
  - *Spotlight Model:* Attention acts like a flashlight beam. The center (focus) sees with high resolution, surrounded by a lower-resolution edge (fringe).
  - *Feature Integration Theory (Treisman):* Basic features (like color or shape) are processed automatically and in parallel across the visual field. But binding those features together to recognize a specific object requires focused, serial attention.
- **Attentional Blindness and Deficits:** Phenomena demonstrating the boundaries and clinical breakdowns of attentional focus.
  - *Inattentional Blindness:* Failing to notice a fully visible, unexpected object because your focus is locked on something else (like the famous invisible gorilla test).
  - *Change Blindness:* Failing to notice a visual change across scene cuts or brief interruptions.
  - *Hemispatial Neglect:* Typically caused by damage to the right posterior parietal cortex. Patients lose the ability to pay attention to the left side of space—eating food only from the right side of a plate, or drawing only the right side of a clock. Their eyes work, but the attentional spotlight cannot look left.
- **Attention vs. Consciousness:** Distinguishing between selective focus and subjective awareness.
  - They are closely linked, but they are not the exact same thing.
  - *Attention without Consciousness:* Invisible stimuli (like subliminal images masked by flashes) can still pull your spatial attention and prime later thoughts.

- *Zombie Behaviors*: Highly practiced motor sequences (like tying shoelaces, balancing on a bike, or adjusting your posture) execute smoothly without needing conscious awareness.
- *Consciousness without Focused Attention*: You can grasp the general "gist" of a scene in as little as 30 milliseconds, far faster than top-down attention can systematically scan the scene.

## Everyday Examples

- **Pop-Out Effect**: If you look at a sea of green circles, a single bright red circle pops out instantly without any effort. But if you have to find a red square hidden among red circles and green squares, you must direct your attention to each shape one by one (conjunction search).
- **Representational Neglect**: When Italian neglect patients were asked to imagine standing in a famous piazza facing the cathedral, they described only the buildings on the right side. When asked to turn around mentally and imagine looking away from the cathedral, they described the buildings they had just ignored, because those buildings were now on their imagined right.

## Key Takeaways

- Attention acts as a bottleneck, filtering the flood of raw sensory input down to what we can consciously handle.
- Basic visual features pop out automatically, but combining features into recognizable objects requires focused attention.
- Hemispatial neglect is an attentional failure, not a sensory vision failure.
- Automatic actions ("zombie behaviors") run smoothly with almost zero conscious attention.

## Glossary

- **Assortative Mating**: When individuals choose partners with similar physical or behavioral traits rather than mating at random.
- **BOLD Effect (Blood Oxygenation Level-Dependent)**: The signal measured by fMRI that tracks the influx of oxygen-rich blood into active brain regions.
- **Central Executive**: The control center of working memory that directs focus and coordinates tasks.
- **Cochlea**: The fluid-filled, snail-shaped structure in the inner ear that converts pressure waves into electrical signals.
- **Dorsal Stream**: The visual pathway leading to the parietal lobe that handles motion and guides "where" and "how" we interact with objects.
- **Genotype**: The full genetic code an organism inherits.
- **Hemispatial Neglect**: An attention disorder (usually after right parietal damage) where a person ignores the opposite (usually left) side of space.
- **Inattentive Blindness**: Failing to see an obvious, unexpected object because attention is focused elsewhere.
- **Phenotype**: The observable physical traits and behaviors produced by an organism's

genotype interacting with the environment.

- **Phoneme:** The smallest distinct sound unit in a language (like the difference between "p" and "b").
- **Phonological Loop:** The verbal component of working memory that temporarily holds speech sounds and keeps them active through mental rehearsal.
- **Prosopagnosia:** A neurological condition where a person cannot recognize familiar faces (face blindness).
- **Retinotopic Map:** A layout in the visual cortex where neighboring neurons represent neighboring points on the retina.
- **Tonotopic Map:** A layout in the ear and auditory cortex where neurons are organized sequentially by sound frequency.
- **Universal Grammar:** Chomsky's theory that humans are born with an innate framework for grammatical rules.
- **Ventral Stream:** The visual pathway leading to the temporal lobe that identifies "what" an object or face is.
- **Visuo-Spatial Sketchpad:** The visual component of working memory used to create and manipulate mental images and spatial layouts.

# Quick Revision Sheet

## Core Ideas

- **Mind & Brain Interdependence:** Cognitive psychology rejects pure biological determinism and pure blank-slate views. Behavior comes from genetic tendencies actively shaping and reacting to environments.
- **Parallel Working Memory:** Memory does not use a single entry bottleneck. Visual and verbal data run through separate channels, managed by an attentional supervisor.
- **Two Visual Highways:** Seeing is split into identification (ventral / what) and physical action/location (dorsal / where/how).
- **Limited Attentional Bandwidth:** Because conscious processing is limited, attention directs neural resources while automated routines run without conscious supervision.

## Key Method Comparisons

| Tool    | What It Measures                                 | Best For...                                                        | Major Limitation                                                        |
|---------|--------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------|
| fMRI    | Blood oxygenation changes (BOLD)                 | Pinpointing active brain regions (high spatial resolution)         | Slow response (poor temporal resolution)                                |
| EEG     | Scalp electrical voltage from firing neurons     | Tracking timing down to the millisecond (high temporal resolution) | Blurry location data (poor spatial resolution)                          |
| PET     | Radioactive tracer tracking glucose / blood flow | Metabolic activity across the brain                                | Involves radiation, highly expensive, slow                              |
| TMS     | Magnetic pulses applied to the scalp             | Testing cause-and-effect by pausing/stimulating brain patches      | Only reaches surface cortex, short-term side effects                    |
| Lesions | Naturally occurring brain injuries               | Linking damaged structures to lost abilities                       | Damage is rarely isolated; no baseline data; plasticity confounds tests |

## Most Important Distinctions

- **Genotype vs. Phenotype:** Genetic blueprint vs. actual real-world expression.
- **Overextension vs. Underextension:** Calling every pet "cat" vs. thinking only your household pet is a "cat".
- **Aphasia vs. Amusia:** Language loss vs. music perception loss (double dissociation).
- **Inattentional Blindness vs. Neglect:** Missing something because you were busy looking at another task vs. an inability to shift attention to one side of space due to brain damage.